

Modular Maths

will start at 7:05 am

$\% \Rightarrow \text{Mod} \rightarrow \text{Remainder}$

$n \% a \Rightarrow \text{remainder if we divide } n/a$

$$10 \% 4 = 2 \rightarrow 4 * 2 + 2 = 10$$

$$13 \% 5 = 3 \rightarrow 5 * 2 + 3 = 13$$

Divident = Quotient * Divisor + Remainder

$$D = q * d + r$$

$$\begin{array}{r} \rightarrow 0 \\ d \leftarrow 5 \mid 13 \rightarrow q \\ \underline{10} \\ 3 \rightarrow r \end{array}$$

Remainder = Divident - Quotient * Divisor
Largest multiple of
divisor $\leq$ Divident

$$\Rightarrow 150 \% 11$$

$$\begin{array}{r} 11 \mid 150 \rightarrow 13 \\ \underline{11} \\ 40 \\ \underline{33} \\ 7 \end{array}$$

Quiz 2: 100 % 7

$$\begin{array}{r} 100 - 98 = 2 \\ - 7 \times 14 \end{array}$$

$$\begin{array}{r} 7 \overline{) 100} 14 \\ \underline{7} \\ 30 \\ \underline{28} \\ 2 \end{array}$$

Q3: Pick the largest #

-43, -42, -76, -35

$$-76 < -43 < -42 < -35$$

Q4: -40 % 7

$$-40 - (-7 \times 5)$$

$$\begin{array}{r} -40 - (-35) \\ -5 \end{array}$$

-5

$$-35 > -40$$

$$\Rightarrow \begin{array}{r} -40 - (-42) \\ 2 \checkmark \end{array}$$

Remainder = Divident - Quotient * Divisor
Largest multiple of
divisor $\leq$ Divident

$$\Rightarrow -60 \% 9$$

$$\times -60 - (-54)$$

$$-54 > -60$$

$$-60 - (-63) \\ = +3$$

$$\boxed{a \% m = m - 1}$$

$$[0 - 8]$$

$$90 \% 9 =$$

$$[0 \rightarrow a - 1]$$

$\Rightarrow$ Python

$$-40 \% 7 = 2$$

$$-60 \% 9 = 3$$

Java / C / C++ / JS

$$-5 + 7 = 2$$

$$-6 + 9 = 3$$

$$\boxed{a \% m + m}$$

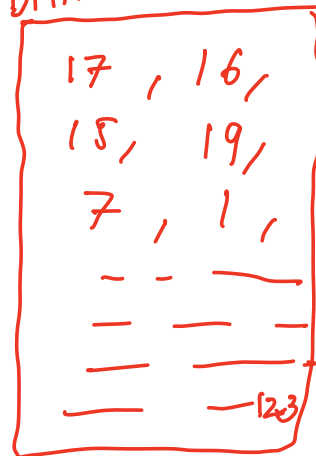
+, -, /, *, %

$$- \infty \quad \swarrow \quad \% 5 \Rightarrow \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$$

+20

find(16) →
find(123) →
126 →

Billions of numbers



Billions

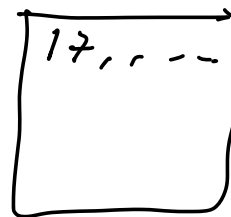
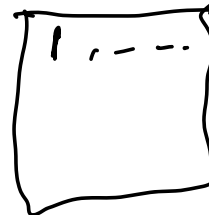
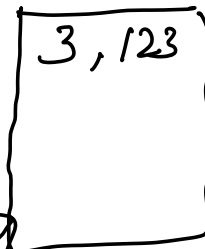
17 % 3

1 % 3

3 % 3

123 % 3

(126) % 3 = 0



2

[1, 100]

1L

17L

- Hashmaps
- Consistency Hashing

Encryption

⇒ Modular properties

$$\underbrace{(a + b) \% m}_{0 - (m-1)} = \underbrace{(a \% m)}_{m-1} + \underbrace{(b \% m)}_{m-1} \% m$$

$$\begin{array}{lcl} a = 6, & b = 8, & m = 10 \\ (6 + 8) \% 10 & & 6 \% 10 + 8 \% 10 \\ 14 \% 10 & & 6 + 8 \\ 4 & & 14 \end{array}$$

$$(a * b) \% m = (a \% m * b \% m) \% m$$

⇒ Implement power function
 $\text{power}(a, n, p) \rightarrow a^n \% p$

$$a = 2, n = 5, p = 7 \rightarrow 2^5 \% 7 = 32 \% 7 = 4$$

$$a = 3, n = 4, p = 6 \rightarrow 3^4 \% 6 = 81 \% 6 = 3$$

```
int power (a, n, b)
{
```

```
    int ans = 1;
```

$$a = 10, n = 40$$

$$ans = 10^{40}$$

↳ int x

```

for (i=1 ; i <= n; i++)
{
    ans = ans * a; → overflow
}
return ans % p;

```

→ long x

```

int power (a, n, b)
{
    long ans = 1;

```

$p = 10^9 + 7$

O(N)

```

    for (i=1 ; i <= n; i++)
    {
        // ans = (ans * a) % p;

        ans = ( ans % p * a % p ) % p
    }
    return ans;
}

```

$p-1$ $p-1$

$$2^4 \% 10$$

$$ans = (1 \% 10 * 2 \% 10) \% 10 = 2$$

$$ans = (2 \% 10 * 2 \% 10) \% 10 = 4$$

$$ans = (4 \% 10 * 2 \% 10) \% 10 = 8$$

$$ans = (8 \% 10 * 2 \% 10) \% 10 = \underline{\underline{6}}$$

$$[(c_{p-1}) * (c_{p-1})] \% p$$

$$[(10^9 + 7 - 1) * (10^9 + 7 - 1)] \% p$$

$$[(10^9 + 6) * (10^9 + 6)] \% p$$

$$\approx 10^{18} \rightarrow \begin{matrix} \text{int} \times \\ \text{long} \end{matrix}$$

Divisibility ruler

Rule for 3 $\Rightarrow$ sum of digits has to be divisible by 3

$$\Rightarrow (4372) \% 3 \Rightarrow (4 + 3 + 7 + 2) \% 3$$

$$\Downarrow$$

$$(4 * 10^3 + 3 * 10^2 + 7 * 10 + 2 * 10^0) \% 3$$

$$[(4 * 10^3) \% 3 + (3 * 10^2) \% 3 + (7 * 10) \% 3 + (2 * 10^0) \% 3]$$

$$(4 \% 3 * 10^3 \% 3) \% 3 +$$

$$\frac{(7 \% 3 * 10 \% 3) \% 3 + (2 \% 3) \% 3}{(7 \% 3) \% 3}$$

$$\overline{(4 \times 3) \div 3}$$

$$\underbrace{(3 \times 3 + 10^2 \times 3) \div 3 +}_{(3 \times 3) \div 3}$$

$$\overline{(2 \times 3) \div 3}$$

$$10 \div 3 = 1$$

$$10^2 \div 3 = 1$$

$$10^3 \div 3 = 1$$

$$\vdots$$

$$10^n \div 3 = 1$$

$$(a \div b) \div b = (a \times b)$$

$$\underbrace{(4 \div 3)}_{0-2} \div 3$$

$$\textcircled{1} \times 3 = 1$$

$$(4 \div 3 + 3 \div 3 + 7 \div 3 + 2 \div 3) \div 3$$

$$(a \div b + b \div b) \div b = (a + b) \div b$$

$$\rightarrow \underline{(4 + 3 + 7 + 2) \div 3}$$

Divisibility by 4

No found by last 2 digits

13824

44621 X

$$(3484) \div 4 \Rightarrow (84) \div 4$$

$$(3 \times 10^3 + 4 \times 10^2 + 8 \times 10 + 4) \div 4$$

$$((3 \times 10^3) \div 4 + (4 \times 10^2) \div 4 + (8 \times 10) \div 4 + 4 \div 4)$$

$$\begin{array}{c}
 \downarrow \qquad \qquad \qquad \downarrow \qquad \qquad \qquad \downarrow \qquad \qquad \qquad \downarrow \\
 \underbrace{(3 \times 4 * 10^3 \times 4) \times 4 +}_{\Downarrow 0} \underbrace{(4 \times 4 * 10^2 \times 4) \times 4 +}_{\Downarrow 0} \underbrace{(0 \times 4 * 10 \times 4) \times 4 +}_{\downarrow \times 4} \underbrace{(4 \times 4) \times 4}_{\times 4}
 \end{array}$$

$$10^0 \times 4 \neq 0$$

$$10^1 \times 4 \neq 0$$

$$10^2 \times 4 = 0$$

$$10^3 \times 4 = 0$$

$$10^4 \times 4 = 0$$

⋮

$$10^n \times 4 = 0$$

$$((0 \times 4 * 10 \times 4) \times 4 + 4 \times 4) \times 4$$

$$((0 * 10) \times 4 + 4 \times 4) \times 4$$

$$((00) \times 4 + 4 \times 4) \times 4$$

$$(00 + 4) \times 4 = 04 \times 4$$

Hw: 5, 8 & 9

Break: 5 min

8: 32

⇒ -40 % 7

$$-40 - (-35) = -5 + m$$

$$40 \% 7$$

$$40 - 35$$

2 complement

00001001

11110110

+1

11110111

→

prints

Javg

-5%

Google

Q = Given a number A in form of array of size N & no. p
Return $A \% p$

$$1 < p \leq 10^9$$

$$1 \leq N \leq 10^5$$

A:

1 2 3 4 4

= 12344

p: 4

$A \% p$

$$12344 \% 4 = 0$$

A:

0 10⁵

$$10^2 \rightarrow 3$$

$$\log \left(\frac{10^{10}}{10^2} \right) \Rightarrow 19 \text{ digits}$$

<

.....

$$\underline{10^{10^7}} = 10^{10000000}$$

$$\begin{aligned} 19 \text{ digits} &\rightarrow 10^{10} \\ 10^5 \text{ digits} &\rightarrow 10^{10^5 - 1} \end{aligned}$$

$$\begin{aligned} & (\overset{0}{3} \quad \overset{1}{8} \quad \overset{2}{4} \quad \overset{3}{3} \quad \overset{4}{6} \quad \overset{5}{0} \quad \overset{6}{9}) \% p \\ & \downarrow \quad \downarrow \\ & \underbrace{(3 \times 10^6) \% p +}_{\downarrow} \underbrace{(8 \times 10^5) \% p +}_{\downarrow} \\ & \underbrace{(3 \% p \times 10^6 \% p) \% p}_{\downarrow} \quad \underbrace{(8 \% p \times 10^5 \% p) \% p}_{\downarrow} \\ & \quad \quad \downarrow \\ & \quad \quad \text{pow}(10, N-1, p) \end{aligned}$$

$$\begin{aligned} A : & \quad \overset{0}{a_0} \quad \overset{1}{a_1} \quad \overset{2}{a_2} \quad \dots \quad \overset{n-2}{a_{n-2}} \quad \overset{n-1}{a_{n-1}} \\ & \quad \downarrow \\ & (a_0 \times 10^{n-1}) \% p + a_1 \times 10^{n-2} \% p - \dots \\ & \quad \downarrow \\ & (\underline{a \% p} \times \text{pow}(10, n-1, p)) \% p \end{aligned}$$

$$\text{ans} = 0;$$

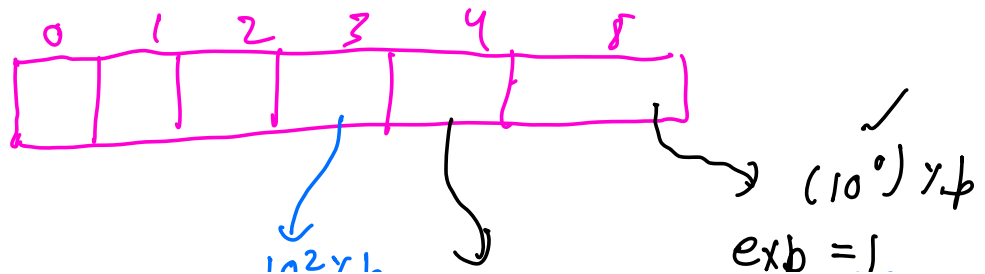
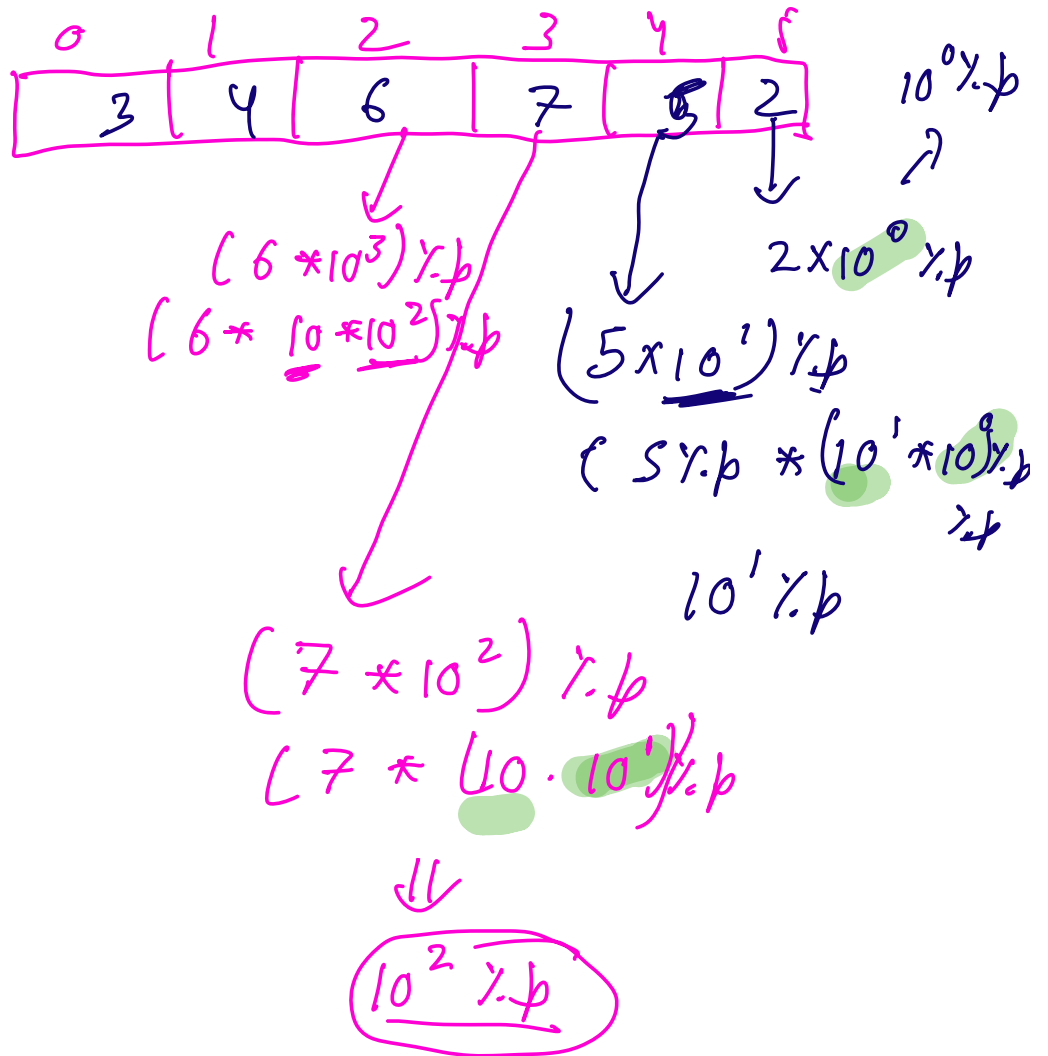
$$\begin{aligned} & \text{for } (\underline{i = 0}; \quad i < \underline{N}; \quad i++) \quad \left. \vphantom{\text{for}} \right\} \rightarrow N \\ & \quad \left\{ \begin{aligned} & \text{ans} += (A[i] \% p \times \text{pow}(10, n-1-i, p)) \% p \\ & \end{aligned} \right. \quad \begin{aligned} & \text{OCN} \\ & \% p \end{aligned} \\ & \quad \left. \vphantom{\text{for}} \right\} \\ & \text{return ans;} \end{aligned}$$

}

TC : $O(N^2)$

SC : $O(1)$

optimize



$$\begin{aligned}
 & \text{exp} = (\text{exp} * 10) \% p \\
 & \text{exp} = 10^2 \% p \\
 & \text{exp} = (10^1 \% p * 10^0 \% p) \% p \\
 & \text{exp} = 10 \% p
 \end{aligned}$$

```

long ans = 0
long exp = 1
for (i = N-1; i >= 0; i--)

```

$O(N)$

```

    ans = (ans * 10 + arr[i]) % p
    exp = (exp * 10) % p
}
return ans;

```

$$(arr[i] \% p * exp \% p) \% p$$

(1 2 3 4)

$$\% 43 \Rightarrow 2$$

$$(1 * 10^3 + 2 * 10^2 + 3 * 10 + 4 * 10^0) \% p$$

$$(1 \% 3 * 1) \% 3$$

$$(3 \% p * 10 \% p) \% p$$

$$(4 \% p * 10^0 \% p) \% p$$

$$exp = 1$$

$$(2 \% p * 1) \% p$$

$$(10 * exp) \% p$$

$$(10 * 1) \% 3$$

$$(4 \% p) \% p$$

$$17.3 \Rightarrow 1$$

$$2$$

$$ans = 0$$

11

$$exp = 1$$

$$(10 * 1) \% 3$$

1

$$\underline{exp = 1}$$

$$(1 * 10) \% 3$$

1